ID	Name	Age	Department	GPA
101	Alice Smith	20	Computer Science	3.85
102	Bob Jones	22	Mathematics	3.40
103	Charlie Brown	19	Physics	3.90
104	Diana Prince	21	Biology	3.75
105	Ethan Hunt	23	Engineering	3.10

Q1. **Create a CSV file given above using Pandas library.**

Answer1.

```
import pandas as pd

data = {

   "id":[101,102,103,104,105]

   "name": ["Alice Smith", "Bob Jones", "Charlie Brown", "Diana Prince",”Ethan Hunt”],

   "age":[20,22,19,21,23]

   "department": [ "Computer Science",  "Mathematics", "Physics", “Biology”, "Engineering”  ],

   "gpa": [3.85, 3.40, 3.90, 3.75, 3.10],

}

df = pd.DataFrame(data)

df.to_csv("student_data.csv", index=False)
```

Q2. **Create a CSV file given above using CSV library**

Answer2.

```
import csv
student_data = [
    ["ID", "Name", "Age", "Department", "GPA"],
    [1, "Alice Smith", 20, "Computer Science", 3.8],
    [2, "Bob Jones", 22, "Mathematics", 3.5],
    [3, "Charlie Brown", 21, "Physics", 3.2],
    [4, "Diana Prince", 23, "Biology", 3.9],
]
with open(“student.csv”, mode="w", newline="", encoding="utf-8") as file:
    writer = csv.writer(file)
    writer.writerows(student_data)
```

Q3. **WAP to replace comma with tab in student.csv file**.

```
import pandas as pd
df = pd.read_csv("student.csv")
df.to_csv("output.tsv", sep="\t", index=False)
print("Done")
```

Q4. **WAP to calculate mean of the GPA of student_data.csv**

Answer4.

```
Import numpy

Import pandas as pd

df = pd.read_csv("student_data.csv")

n=numpy.mean(df.gpa)

print(n)
```

Q5. **WAP to calculate the mode of the GPA of student_data.csv**

Answer 5.

```
from scipy import stats as st

Import pandas as pd

df = pd.read_csv("student_data.csv")

r=st.mode(df.gpa)

print(r)
```

Q6. **WAP to display the scatter graph of student_data.csv file with id and gpa.**

Answer6.

```
import matplotlib.pyplot as plt

Import pandas as pd

df = pd.read_csv("student_data.csv")

plt.scatter(df.id, df.gpa)

plt.title("Scatter Plot")

plt.xlabel("ID")

plt.ylabel("GPA")

plt.show()
```

Q7. **WAP to plot a box plot for gpa column of student_data.csv**

```
import matplotlib.pyplot as plt

Import pandas as pd

df = pd.read_csv("student_data.csv")

plt.boxplot(df.gpa)

plt.xlabel("Groups")

plt.ylabel("GPA")

plt.title("Box Plot")

plt.show()
```

Q8. **WAP to add a new row in the student_data.csv file with data as [106,”Tom”,24,”Chemistry”,””]**

Answer 8.

```
import csv

new_row[106,”Tom”,24,”Chemistry”,””]

with open('student_data.csv', mode='a',newline='') as file:

    writer = csv.writer(file)

    writer.writerow(new_row)
```

Q9. **WAP to delete all the missing values in student_data.csv file.**

```
import pandas as pd

data=pd.read_csv('student_data.csv')

data1=data.dropna()

print(data1)

data1.to_csv(‘student_data.csv', index=False)
```

Q10. **WAP to replace the missing values in student_data.csv file with mean of the gpa.**

```
import pandas as pd

data=pd.read_csv('student_data.csv')

data['gpa']=data['gpa'].fillna(data['gpa'].mean())

print(data)

data1.to_csv(‘student_data.csv', index=False)
```

Q11. **WAP to display only first 2 rows and last 1 row and complete information of student_data.csv file.**

```
import pandas as pd

df= pd.read_csv("student_data.csv")

print(df.head(2))

print(df.tail(1))

df.info()
```